

AI Transit Pipeline

Installation Guide

Version 2.0 · 2026-08-31

Applicable to: `fetch_repo.sh` · `scan_pipeline.sh` · `ai_transit.sh` · `generate_excel_report.py` · `selfcheck.py`

Table of Contents

- Overview
- Prerequisites
- Core System Packages
- Python Environment
- Tool-by-Tool Installation
 - 5.1 betterleaks — secret detection (L1)
 - 5.2 detect-secrets — entropy scanning (L1)
 - 5.3 ClamAV — antivirus (L1)
 - 5.4 YARA — custom IOC rules (L1)
 - 5.5 Semgrep — OWASP / CWE / CERT (L2)
 - 5.6 trivy — SCA / CVE (L3)
 - 5.7 pip-audit — Python CVE (L3)
 - 5.8 safety — Python advisories (L3)
 - 5.9 npm + npm audit — Node.js CVE (L3)
 - 5.10 Bandit — Python SAST (L5)
 - 5.11 ShellCheck — shell SAST (L5)
 - 5.12 cppcheck — C/C++ SAST (L5)
 - 5.13 hadolint — Dockerfile linter (L5)
 - 5.14 checkov — Terraform / IaC (L5)
 - 5.15 ScanCode Toolkit — licence & copyright (L6)
- Pipeline Scripts Installation
- Directory Structure Setup
- Environment Variables & Flags
- Full Installation Verification
- Air-Gapped Operation
- Sample Scans — Testing the Pipeline
- Self-Scan: Verifying the Installation is Safe
- Security Hardening Recommendations
- Troubleshooting
- Running the Test Suite
- Continuous Integration
- Docker Image — Build Arguments & Integrity

1. Overview

The AI Transit Pipeline is a 6-layer security gateway that scans AI-generated code before enterprise import.

Layer	Tools	Blocks on
L1 — Secrets & AV	betterleaks, detect-secrets, ClamAV, YARA	Secret/credential/malware detected
L2 — OWASP/CWE	Semgrep	ERROR/WARNING severity finding
L3 — SCA/CVE	trivy, pip-audit, safety, npm audit	HIGH/CRITICAL CVE in dependency
L4 — Patterns	grep (built-in)	CWE-798/22/918/327/338 match
L5 — Per-type SAST	Bandit, ShellCheck, cppcheck, hadolint, checkov, Semgrep per-lang	Tool-specific finding (also covers Rust, Kotlin, C#)

Layer	Tools	Blocks on
L6 — Licence	ScanCode Toolkit	CRITICAL/HIGH CVE in detected package

Verdict: any FAIL in any layer → package quarantined. WARNs are logged but do not block.

2. Prerequisites

Supported platforms: Ubuntu 22.04 LTS / Debian 12 (production), Ubuntu 24.04 LTS (recommended).

Windows users: install via **WSL2** (see Section 4 of base guide).

Minimum hardware: 4-core CPU, 8 GB RAM, 20 GB free disk.

3. Core System Packages

Install the baseline packages first. These are required for the pipeline to run at all.

```
sudo apt-get update && sudo apt-get upgrade -y

sudo apt-get install -y \
bash git curl ca-certificates wget gpg \
jq zip unzip file coreutils \
python3 python3-pip python3-venv \
build-essential golang-go \
nodejs npm
```

Verify:

```
bash --version | head -1 # expect: GNU bash, version 5.x
git --version # expect: git version 2.x
python3 --version # expect: Python 3.10+
go version # expect: go1.21+
node --version # expect: v18+ or v20+
jq --version # expect: jq-1.6+
```

4. Python Virtual Environment

Using a dedicated venv avoids conflicts with system packages and makes the installation self-contained.

```
python3 -m venv /opt/ai-transit/venv
echo 'source /opt/ai-transit/venv/bin/activate' >> ~/.bashrc
source /opt/ai-transit/venv/bin/activate
pip install --upgrade pip
```

All pip install commands in this guide assume the venv is active.

Verify:

```
which python3 # must show /opt/ai-transit/venv/bin/python3
python3 --version # Python 3.10+
```

5. Tool-by-Tool Installation

Each section covers: finding the latest version, installing, verifying, offline setup, and a functional test.

5.1 betterleaks — Secret Detection (Layer 1) {#betterleaks}

What it does: scans all file types for secrets, API keys, credentials and tokens. Successor to gitleaks with a more expressive allowlist system.

Find the latest version

```
# Query GitHub API for latest release tag
curl -s https://api.github.com/repos/betterleaks/betterleaks/releases/latest \
| jq -r '.tag_name'
```

Install (Go — recommended, produces a static binary)

```
go install github.com/betterleaks/betterleaks@latest
sudo cp ~/go/bin/betterleaks /usr/local/bin/betterleaks
```

Install (Homebrew — macOS / Linux with brew)

```
brew install betterleaks
```

Install (Fedora / RHEL)

```
sudo dnf install betterleaks
```

Verify installation

```
betterleaks --version
# Expected output: betterleaks version x.y.z
```

Offline operation

betterleaks has **no external database** — all detection rules are embedded in the binary. Once installed, it works fully offline.

Functional test

```
# Create a test file with a fake secret
mkdir /tmp/bl-test
echo 'AWS_SECRET_ACCESS_KEY="wJa1rXUtnFEMI/K7MDENG/bPxRfiCYEXAMPLEKEY"' \
> /tmp/bl-test/config.py

betterleaks dir /tmp/bl-test -v
# Expected: non-zero exit code, finding reported for config.py

# Clean file — should PASS
echo 'import os; key = os.environ["AWS_SECRET_ACCESS_KEY"]' \
> /tmp/bl-test/config_clean.py
```

```
betterleaks dir /tmp/bl-test/config_clean.py -v
# Expected: exit code 0

rm -rf /tmp/bl-test
```

5.2 detect-secrets — Entropy Scanning (Layer 1) {#detect-secrets}

What it does: complements betterleaks with Shannon entropy analysis and regex-based patterns to find high-entropy strings that look like secrets.

Find the latest version

```
pip index versions detect-secrets 2>/dev/null | head -1
# or:
curl -s https://pypi.org/pypi/detect-secrets/json | jq -r '.info.version'
```

Install

```
pip install detect-secrets
```

Verify installation

```
detect-secrets --version
# Expected: x.y.z
```

Offline operation

No external database. Works fully offline once installed.

Functional test

```
mkdir /tmp/ds-test
echo 'password = "Sup3rS3cr3tP@ssw0rd!"' > /tmp/ds-test/app.py

detect-secrets scan /tmp/ds-test/app.py
# Expected: JSON with non-empty "results" containing app.py

echo 'password = os.getenv("PASSWORD")' > /tmp/ds-test/app_clean.py
detect-secrets scan /tmp/ds-test/app_clean.py
# Expected: JSON with empty "results" {}

rm -rf /tmp/ds-test
```

5.3 ClamAV — Antivirus (Layer 1) {#clamav}

What it does: scans files against a database of known malware signatures. Requires regular DB updates to stay effective.

Find the latest version

```
apt-cache policy clamav | grep Candidate
# or for the latest upstream:
curl -s https://www.clamav.net/downloads | grep -oP 'clamav-[0-9.]+(=?\.tar)' | head -1
```

Install

```
sudo apt-get install -y clamav clamav-daemon clamav-freshclam
```

Update virus database (required before first use)

```
sudo systemctl stop clamav-freshclam
sudo freshclam
sudo systemctl start clamav-freshclam
sudo systemctl enable clamav-freshclam
```

Verify installation

```
clamscan --version
# Expected: ClamAV x.y.z/NNNNN/...
freshclam --version
# Expected: ClamAV x.y.z
```

Offline operation

ClamAV works offline once the virus DB is downloaded. To prepare for offline use:

```
# On a machine with internet access, download the DB files
sudo freshclam

# The DB files are stored at (copy these to the offline machine):
ls /var/lib/clamav/
# main.cvd (or main.cld), daily.cvd (or daily.cld), bytecode.cvd

# On the offline machine, copy the DB files to /var/lib/clamav/
# then reload:
sudo systemctl restart clamav-daemon
```

Functional test

```
# ClamAV includes the EICAR test signature – a safe test virus string
echo 'X50!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*' \
> /tmp/eicar.txt

clamscan /tmp/eicar.txt
# Expected: /tmp/eicar.txt: Eicar-Signature FOUND
# Expected exit code: 1

clamscan /bin/ls
# Expected: /bin/ls: OK
# Expected exit code: 0

rm /tmp/eicar.txt
```

5.4 YARA — Custom IOC Rules (Layer 1) {#yara}

What it does: loads organisation-specific YARA rules from yara-rules/ and matches them against every file. Fully customisable pattern engine.

Find the latest version

```
apt-cache policy yara | grep Candidate
```

```
# or from GitHub:
curl -s https://api.github.com/repos/VirusTotal/yara/releases/latest \
| jq -r '.tag_name'
```

Install

```
sudo apt-get install -y yara
```

Verify installation

```
yara --version
# Expected: x.y.z
```

Offline operation

YARA has no external database — all rules are local .yar files. Works fully offline.

Create a minimal test rule

```
mkdir -p /opt/ai-transit/yara-rules

cat > /opt/ai-transit/yara-rules/test.yar << 'EOF'
rule Suspicious_Base64_Exec {
meta:
description = "Detects base64-encoded exec calls"
strings:
$b64_exec = /eval\(base64_decode\(/ nocase
condition:
$b64_exec
}
EOF
```

Functional test

```
echo '<?php eval(base64_decode("dw5saw5r...")); ?>' > /tmp/test.php

yara /opt/ai-transit/yara-rules/test.yar /tmp/test.php
# Expected: Suspicious_Base64_Exec /tmp/test.php

echo '<?php echo "Hello World"; ?>' > /tmp/clean.php
yara /opt/ai-transit/yara-rules/test.yar /tmp/clean.php
# Expected: (no output – no match)

rm /tmp/test.php /tmp/clean.php
```

5.5 Semgrep — OWASP / CWE / CERT Static Analysis (Layer 2) {#semgrep}

What it does: runs four security rulesets (OWASP Top 10, CWE Top 25, security-audit, secrets) across all supported languages using pattern matching on the AST.

Find the latest version

```
curl -s https://pypi.org/pypi/semgrep/json | jq -r '.info.version'
```

Install

```
pip install semgrep
```

Verify installation

```
semgrep --version  
# Expected: 1.x.x
```

Offline operation

Semgrep needs internet access **only on the first run** to download rules. Prepare for offline:

```
# On a machine with internet, pre-download rules to a local directory  
mkdir -p /opt/ai-transit/semgrep-rules  
semgrep --config p/owasp-top-ten --dump-config > /opt/ai-transit/semgrep-rules/owasp.yaml  
semgrep --config p/cwe-top-25 --dump-config > /opt/ai-transit/semgrep-rules/cwe.yaml  
semgrep --config p/security-audit --dump-config > /opt/ai-transit/semgrep-rules/audit.yaml  
semgrep --config p/secrets --dump-config > /opt/ai-transit/semgrep-rules/secrets.yaml  
  
# On the offline machine, run against local rules:  
semgrep --config /opt/ai-transit/semgrep-rules/ <repo_dir>
```

Functional test

```
cat > /tmp/test_sqli.py << 'EOF'  
import sqlite3  
def get_user(username):  
    conn = sqlite3.connect("db.sqlite3")  
    query = "SELECT * FROM users WHERE name = '" + username + "'"   
    return conn.execute(query).fetchall()  
EOF
```

```
semgrep --config p/owasp-top-ten --json /tmp/test_sqli.py \  
| jq '.results | length'  
# Expected: 1 or more (SQL injection detected)
```

```
cat > /tmp/test_clean.py << 'EOF'  
import sqlite3  
def get_user(username):  
    conn = sqlite3.connect("db.sqlite3")  
    return conn.execute("SELECT * FROM users WHERE name = ?", (username,)).fetchall()  
EOF
```

```
semgrep --config p/owasp-top-ten --json /tmp/test_clean.py \  
| jq '.results | length'  
# Expected: 0
```

```
rm /tmp/test_sqli.py /tmp/test_clean.py
```

5.6 trivy — SCA / CVE (Layer 3) {#trivy}

What it does: scans dependency manifests (requirements.txt, package-lock.json, go.sum, pom.xml, Cargo.lock ...) against NVD, OSV and GitHub Advisory databases.

Find the latest version


```
curl -s https://api.github.com/repos/aquasecurity/trivy/releases/latest \
| jq -r '.tag_name'
```

Install (official script — recommended)

```
curl -sL https://raw.githubusercontent.com/aquasecurity/trivy/main/contrib/install.sh \
| sudo sh -s -- -b /usr/local/bin
```

Install (apt repository)

```
sudo apt-get install -y wget gnupg
wget -qO - https://aquasecurity.github.io/trivy-repo/deb/public.key \
| sudo gpg --dearmor -o /usr/share/keyrings/trivy.gpg
echo "deb [signed-by=/usr/share/keyrings/trivy.gpg] \
https://aquasecurity.github.io/trivy-repo/deb generic main" \
| sudo tee /etc/apt/sources.list.d/trivy.list
sudo apt-get update && sudo apt-get install -y trivy
```

Verify installation

```
trivy --version
# Expected: Version: x.y.z
```

Download vulnerability database (required before first offline use)

```
# Pre-download the CVE database (requires internet)
trivy image --download-db-only
trivy image --download-java-db-only

# DB is cached at:
ls ~/.cache/trivy/db/
```

Offline operation

```
# After downloading the DB, run offline:
trivy fs <repo_dir> \
--scanners vuln \
--severity HIGH,CRITICAL \
--skip-db-update \
--offline-scan
```

Functional test

```
mkdir /tmp/trivy-test

# Create a requirements.txt with a known vulnerable package
cat > /tmp/trivy-test/requirements.txt << 'EOF'
Pillow==9.0.0
requests==2.18.0
EOF

trivy fs /tmp/trivy-test \
--scanners vuln \
--severity HIGH,CRITICAL \
--format table
```

```
# Expected: findings for known CVEs in Pillow 9.0.0 and requests 2.18.0

# Clean requirements
cat > /tmp/trivy-test/requirements_clean.txt << 'EOF'
Pillow==10.3.0
requests==2.32.3
EOF

trivy fs /tmp/trivy-test/requirements_clean.txt \
--scanners vuln \
--severity HIGH,CRITICAL \
--format table
# Expected: No vulnerabilities found

rm -rf /tmp/trivy-test
```

5.7 pip-audit — Python CVE (Layer 3) {#pip-audit}

What it does: audits Python requirements files against the Python Packaging Advisory Database (PyPA) and OSV.

Find the latest version

```
curl -s https://pypi.org/pypi/pip-audit/json | jq -r '.info.version'
```

Install

```
pip install pip-audit
```

Verify installation

```
pip-audit --version
# Expected: pip-audit x.y.z
```

Offline operation

pip-audit requires internet to query the PyPA/OSV database. For offline use, use trivy with `--offline-scan` instead (covers the same packages).

Functional test

```
cat > /tmp/req_vuln.txt << 'EOF'
Pillow==9.0.0
requests==2.18.0
EOF

pip-audit -r /tmp/req_vuln.txt
# Expected: one or more vulnerabilities listed, exit code 1

cat > /tmp/req_clean.txt << 'EOF'
Pillow==10.3.0
requests==2.32.3
EOF

pip-audit -r /tmp/req_clean.txt
# Expected: No known vulnerabilities found, exit code 0
```

```
rm /tmp/req_vuln.txt /tmp/req_clean.txt
```

5.8 safety — Python Advisories (Layer 3) {#safety}

What it does: checks Python dependencies against the Safety DB (PyUp.io), a curated advisory database with additional entries not always in OSV.

Find the latest version

```
curl -s https://pypi.org/pypi/safety/json | jq -r '.info.version'
```

Install

```
pip install safety
```

Verify installation

```
safety --version  
# Expected: safety, version x.y.z
```

Offline operation

safety requires internet for DB queries. For offline environments, use trivy as the primary SCA scanner.

Functional test

```
cat > /tmp/req_vuln.txt << 'EOF'  
Pillow==9.0.0  
EOF  
  
safety check -r /tmp/req_vuln.txt  
# Expected: vulnerabilities found, exit code 64  
  
cat > /tmp/req_clean.txt << 'EOF'  
Pillow==10.3.0  
EOF  
  
safety check -r /tmp/req_clean.txt  
# Expected: No known security vulnerabilities found, exit code 0  
  
rm /tmp/req_vuln.txt /tmp/req_clean.txt
```

5.9 npm + npm audit — Node.js CVE (Layer 3) {#npm-audit}

What it does: built into npm — audits package-lock.json against the npm security advisory registry for known CVEs in Node.js dependencies.

Find the latest version

```
npm --version  
node --version  
# npm is bundled with Node.js; update both together
```

Install (via NodeSource — LTS)

```
curl -fsSL https://deb.nodesource.com/setup_lts.x | sudo -E bash -  
sudo apt-get install -y nodejs
```

Verify installation

```
node --version # Expected: v20.x.x or v22.x.x  
npm --version # Expected: 10.x.x or later  
npm audit --version 2>/dev/null || echo "npm audit is built-in"
```

Offline operation

npm audit requires internet to query the npm advisory registry. No offline mode is available.

Functional test

```
mkdir /tmp/npm-test && cd /tmp/npm-test  
  
# Create a package.json with a known vulnerable package  
cat > package.json << 'EOF'  
{  
  "name": "test",  
  "version": "1.0.0",  
  "dependencies": {  
    "lodash": "4.17.4"  
  }  
}  
EOF  
  
npm install --package-lock-only 2>/dev/null  
npm audit --json | jq '.metadata.vulnerabilities'  
# Expected: object with non-zero high/critical counts  
  
cd / && rm -rf /tmp/npm-test
```

5.10 Bandit — Python SAST (Layer 5) {#bandit}

What it does: Python-specific static analyser with 100+ plugins detecting SQL injection, shell injection, hardcoded passwords, weak crypto, insecure deserialization, and more.

Find the latest version

```
curl -s https://pypi.org/pypi/bandit/json | jq -r '.info.version'
```

Install

```
pip install bandit
```

Verify installation

```
bandit --version  
# Expected: bandit x.y.z (Python x.y.z)
```

Offline operation

Bandit has no external database. Works fully offline.

Functional test

```

cat > /tmp/test_bandit.py << 'EOF'
import subprocess
import hashlib

# B602 - subprocess with shell=True
def run(cmd):
    subprocess.call(cmd, shell=True)

# B324 - use of weak hash
def hash_password(pwd):
    return hashlib.md5(pwd.encode()).hexdigest()
EOF

bandit -ll /tmp/test_bandit.py
# Expected: Issue: [B602] ... [B324] ... exit code 1

cat > /tmp/test_bandit_clean.py << 'EOF'
import subprocess
import hashlib

def run(cmd_list):
    subprocess.call(cmd_list)

def hash_password(pwd):
    return hashlib.sha256(pwd.encode()).hexdigest()
EOF

bandit -ll /tmp/test_bandit_clean.py
# Expected: No issues identified, exit code 0

rm /tmp/test_bandit.py /tmp/test_bandit_clean.py

```

5.11 ShellCheck — Shell Script SAST (Layer 5) {#shellcheck}

What it does: static analyser for bash/sh/dash/ksh. Detects quoting errors, command injection risks, undefined variables, deprecated syntax, and unsafe patterns.

Find the latest version

```

apt-cache policy shellcheck | grep Candidate
# or latest binary from GitHub:
curl -s https://api.github.com/repos/koalaman/shellcheck/releases/latest \
| jq -r '.tag_name'

```

Install (apt)

```

sudo apt-get install -y shellcheck

```

Install (latest binary from GitHub)

```

VERSION=$(curl -s https://api.github.com/repos/koalaman/shellcheck/releases/latest \
| jq -r '.tag_name')
curl -sSL "https://github.com/koalaman/shellcheck/releases/download/${VERSION}/shellcheck-${VERSION}.linux.x86_64.tar.xz" \

```

```
| tar -xJ --strip-components=1 -C /tmp
sudo mv /tmp/shellcheck /usr/local/bin/shellcheck
```

Verify installation

```
shellcheck --version
# Expected: version: x.y.z
```

Offline operation

ShellCheck has no external database. Works fully offline.

Functional test

```
cat > /tmp/test.sh << 'EOF'
#!/bin/bash
filename=$1
if [ $filename == "admin" ]; then
echo "Hello $filename"
fi
rm -rf /$filename
EOF

shellcheck --severity=warning /tmp/test.sh
# Expected: SC2086 (unquoted variable), SC2115 (unsafe rm), exit code 1

cat > /tmp/test_clean.sh << 'EOF'
#!/bin/bash
filename="$1"
if [[ "$filename" == "admin" ]]; then
echo "Hello $filename"
fi
EOF

shellcheck --severity=warning /tmp/test_clean.sh
# Expected: exit code 0

rm /tmp/test.sh /tmp/test_clean.sh
```

5.12 cppcheck — C/C++ SAST (Layer 5) {#cppcheck}

What it does: static analyser for C and C++ detecting buffer overflows, memory leaks, null-pointer dereferences, use-after-free, and undefined behaviour without compiling.

Find the latest version

```
apt-cache policy cppcheck | grep Candidate
# or from GitHub:
curl -s https://api.github.com/repos/danmar/cppcheck/releases/latest \
| jq -r '.tag_name'
```

Install (apt)

```
sudo apt-get install -y cppcheck
```

Verify installation

```
cppcheck --version
# Expected: Cppcheck x.y
```

Offline operation

cppcheck has no external database. Works fully offline.

Functional test

```
cat > /tmp/test.cpp << 'EOF'
#include <string.h>
void test() {
char buf[10];
strcpy(buf, "Hello, this string is too long and will overflow the buffer!");
int* p = new int(42);
delete p;
*p = 100; // use after free
}
EOF

cppcheck --enable=warning,security --error-exitcode=1 /tmp/test.cpp
# Expected: errors for buffer overflow and use-after-free, exit code 1

cat > /tmp/test_clean.cpp << 'EOF'
#include <cstring>
void test() {
char buf[100];
strncpy(buf, "Hello", sizeof(buf) - 1);
buf[sizeof(buf) - 1] = '\0';
}
EOF

cppcheck --enable=warning,security --error-exitcode=1 /tmp/test_clean.cpp
# Expected: no errors, exit code 0

rm /tmp/test.cpp /tmp/test_clean.cpp
```

5.13 hadolint — Dockerfile Linter (Layer 5) {#hadolint}

What it does: enforces Dockerfile best practices and detects security misconfigurations: running as root, :latest tags, baked-in secrets, ADD vs COPY, shell injection.

Find the latest version

```
curl -s https://api.github.com/repos/hadolint/hadolint/releases/latest \
| jq -r '.tag_name'
```

Install (binary)

```
VERSION=$(curl -s https://api.github.com/repos/hadolint/hadolint/releases/latest \
| jq -r '.tag_name')
curl -sSL "https://github.com/hadolint/hadolint/releases/download/${VERSION}/hadolint-Linux-x86_64" \
-o /usr/local/bin/hadolint
sudo chmod +x /usr/local/bin/hadolint
```

Verify installation

```
hadolint --version
# Expected: Haskell Dockerfile Linter x.y.z
```

Offline operation

hadolint has no external database. Works fully offline.

Functional test

```
cat > /tmp/Dockerfile_bad << 'EOF'
FROM ubuntu:latest
RUN apt-get update && apt-get install -y curl
ADD https://example.com/script.sh /tmp/
ENV PASSWORD=mysecretpassword
RUN /tmp/script.sh
EOF

hadolint /tmp/Dockerfile_bad
# Expected: DL3007 (latest tag), DL3009 (delete apt cache), SC2046, etc.

cat > /tmp/Dockerfile_good << 'EOF'
FROM ubuntu:22.04
RUN apt-get update \
&& apt-get install -y --no-install-recommends curl \
&& rm -rf /var/lib/apt/lists/*
COPY script.sh /tmp/script.sh
USER nobody
EOF

hadolint /tmp/Dockerfile_good
# Expected: no output or only style suggestions, exit code 0

rm /tmp/Dockerfile_bad /tmp/Dockerfile_good
```

5.14 checkov — Terraform / IaC SAST (Layer 5) {#checkov}

What it does: static analysis for Terraform, CloudFormation, Kubernetes YAML, Ansible and ARM templates. Maps findings to CIS Benchmarks, NIST, SOC2, and OWASP. 1000+ built-in checks.

Find the latest version

```
curl -s https://pypi.org/pypi/checkov/json | jq -r '.info.version'
```

Install

```
pip install checkov
```

Verify installation

```
checkov --version
# Expected: x.y.z
```

Offline operation

checkov uses local checks only — works fully offline. No external DB required.

Functional test

```
mkdir /tmp/tf-test

cat > /tmp/tf-test/main.tf << 'EOF'
resource "aws_s3_bucket" "public_bucket" {
  bucket = "my-public-bucket"
  acl = "public-read"
}

resource "aws_security_group" "open" {
  ingress {
    from_port = 22
    to_port = 22
    protocol = "tcp"
    cidr_blocks = ["0.0.0.0/0"]
  }
}
EOF

checkov -d /tmp/tf-test --framework terraform --compact
# Expected: FAILED checks (public S3 bucket, open SSH to world)

cat > /tmp/tf-test/main_secure.tf << 'EOF'
resource "aws_s3_bucket" "private_bucket" {
  bucket = "my-private-bucket"
}
resource "aws_s3_bucket_acl" "private" {
  bucket = aws_s3_bucket.private_bucket.id
  acl = "private"
}
EOF

checkov -f /tmp/tf-test/main_secure.tf --framework terraform --compact
# Expected: all checks passed or significantly fewer failures

rm -rf /tmp/tf-test
```

5.15 ScanCode Toolkit — Licence & Copyright (Layer 6) {#scancode}

What it does: full-text licence detection using 30 000+ licence texts (SPDX), copyright notice extraction, package manifest detection, and CVE lookup in detected packages. Produces JSON/SPDX/CycloneDX reports.

Find the latest version

```
curl -s https://pypi.org/pypi/scancode-toolkit/json | jq -r '.info.version'
```

Install

```
pip install scancode-toolkit
```

ScanCode is a large package (~500 MB installed). Installation may take 5–10 minutes.

Verify installation

```
scancode --version
# Expected: ScanCode version x.y.z
```

Offline operation

ScanCode works **fully offline** — all licence texts and detection logic are bundled in the package. No external DB or internet required at scan time.

Functional test

```
mkdir /tmp/sc-test

# File with a GPL licence header
cat > /tmp/sc-test/app.py << 'EOF'
# This program is free software: you can redistribute it and/or modify
# it under the terms of the GNU General Public License as published by
# the Free Software Foundation, either version 3 of the License, or
# (at your option) any later version.
# Copyright (C) 2024 Example Corp.
import os

def main():
    print("hello")
EOF

# File with a permissive MIT licence
cat > /tmp/sc-test/lib.py << 'EOF'
# MIT License
# Copyright (c) 2024 Example Corp
# Permission is hereby granted, free of charge, to any person obtaining a copy...
def helper():
    pass
EOF

scancode --license --copyright \
--json-pp /tmp/sc-report.json \
--quiet \
/tmp/sc-test

# Check detections
jq '.files[] | {path: .path, licenses: [.license_detections[].matches[].spdx_license_expression]}' \
/tmp/sc-report.json

# Expected: GPL-3.0-or-later detected in app.py, MIT in lib.py

rm -rf /tmp/sc-test /tmp/sc-report.json
```

6. Pipeline Scripts Installation

6.1 Clone from GitHub

```
git clone https://github.com/gaillotte/claude.git
cd claude
git checkout claude/vigilant-carson-f8twy0
```

6.2 Install Python dependencies for report generation

```
pip install openpyxl reportlab
```

6.3 Make scripts executable

```
chmod +x ai_transit.sh fetch_repo.sh scan_pipeline.sh
```

6.4 Verify scripts

```
bash -n ai_transit.sh && echo "ai_transit.sh: syntax OK"
bash -n fetch_repo.sh && echo "fetch_repo.sh: syntax OK"
bash -n scan_pipeline.sh && echo "scan_pipeline.sh: syntax OK"
python3 -c "import ast; ast.parse(open('generate_excel_report.py').read())" \
&& echo "generate_excel_report.py: syntax OK"
python3 -c "import ast; ast.parse(open('selfcheck.py').read())" \
&& echo "selfcheck.py: syntax OK"
```

6.5 Run the test suite

This is the fastest way to confirm the installation is sound. It needs no scanning tools — see §15.

```
./tests/run_tests.sh
# Expected: ✓ 51/51 passed
```

6.6 Generate the integrity manifest

```
python3 selfcheck.py --write-manifest
```

Do this once the bundle is in its final location. See §12.2 for why the manifest is generated at install time rather than shipped in version control.

7. Directory Structure Setup

```
sudo mkdir -p /opt/ai-transit/{fetch,quarantine,approved,reports,logs,yara-rules}
sudo chmod 700 /opt/ai-transit/quarantine
sudo chown -R $USER:$USER /opt/ai-transit
mkdir -p Good
```

Copy your YARA rules to `/opt/ai-transit/yara-rules/` (see §5.4 for a minimal test rule).

8. Environment Variables & Flags

8.1 Command-line flags

```
./ai-transit.sh [FLAGS] <repo_url_or_path> [branch]

--quiet Suppress all output except the final PASS/FAIL verdict (CI mode)
--verbose Full debug output including per-file scan details
--min-severity LEVEL Severity threshold: low | medium | high (default) | critical
Findings below the threshold are logged as WARN, not FAIL
--since COMMIT Diff mode: only scan files changed since this commit SHA
--report-only Always exit 0 even on FAIL (observation / audit mode).
Also leaves the fetched repo in place instead of
quarantining it, so nothing is moved or deleted.
--no-zip Skip creation of the approved ZIP archive
--no-excel Skip generation of the Excel report
--offline Air-gapped mode: no scanner attempts a network call.
Uses locally staged rules and databases, and reports any
layer that could not run. See section 10.
```

--no-zip --no-excel is the usual combination for CI, where the JSON and HTML reports are consumed by another job and the archive would only be discarded.

8.2 Environment variables

Variable	Default	Description
WORK_DIR	/opt/ai-transit	Root working directory
OUTPUT_DIR	<script_dir>/Good	Output directory for approved ZIPs
GITHUB_TOKEN	(unset)	Personal access token for private GitHub repos
MAX_SIZE_MB	500	Repository size limit in MB
MIN_SEVERITY	high	Minimum severity to block (low\ medium\ high\ critical)
VERBOSITY	normal	Log verbosity passed to scanner (quiet\ normal\ verbose)
SINCE_COMMIT	(unset)	Diff mode commit SHA (same as --since)
OFFLINE	false	Air-gapped mode (same as --offline)
OFFLINE_CACHE	\$WORK_DIR/offline-cache	Root of staged offline assets
SEMGREP_RULES_DIR	\$OFFLINE_CACHE/semgrep-rules	Exported Semgrep rulesets
TRIVY_CACHE_DIR	\$OFFLINE_CACHE/trivy-db	trivy vulnerability database
CLAMAV_DB_DIR	\$OFFLINE_CACHE/clamav	ClamAV signature files

Add persistent values to ~/.bashrc:

```
export WORK_DIR="/opt/ai-transit"
export OUTPUT_DIR="/opt/ai-transit/approved"
```

Or pass inline:

```
WORK_DIR=/data/ai-transit ./ai-transit.sh https://github.com/org/repo
```

8.3 Per-repo exclusions

Place either of these files in the **root of the scanned repository** (not the pipeline directory):

File	Format	Effect
.transitignore	gitignore-style patterns	Files matching patterns are excluded from all scan layers
.transit-allow.json	JSON array of {rule, path, reason}	Matching FAIL findings are downgraded to WARN

.transit-allow.json **example:**

```
[
{
  "rule": "CWE-798",
  "path": "tests/fixtures/dummy_key.py",
  "reason": "Test fixture – not a real credential"
},
{
  "rule": "binary",
  "path": "assets/logo.png",
  "reason": "Approved binary asset"
}
]
```

9. Full Installation Verification

Save this as verify_install.sh and run it after completing all sections above:

```
#!/usr/bin/env bash
# verify_install.sh – checks all required and optional tools

PASS=0; WARN=0; FAIL=0

check() {
  local name="$1" cmd="$2"
  if command -v "$cmd" &>/dev/null; then
    echo -e "\033[32m[OK]\033[0m $name"
    (( PASS++ ))
  else
    echo -e "\033[33m[WARN]\033[0m $name – optional, not installed"
    (( WARN++ ))
  fi
}
```

```

fi
}

require() {
    local name="$1" cmd="$2"
    if command -v "$cmd" &>/dev/null; then
        echo -e "\033[32m[OK]\033[0m $name"
        (( PASS++ ))
    else
        echo -e "\033[31m[FAIL]\033[0m $name - REQUIRED, not installed"
        (( FAIL++ ))
    fi
}

echo "=== Core system ==="
require "bash" bash
require "git" git
require "python3" python3
require "pip" pip
require "curl" curl
require "jq" jq
require "zip" zip
require "sha256sum" sha256sum
require "file" file

echo ""
echo "=== Python report packages ==="
python3 -c "import openpyxl" 2>/dev/null \
&& { echo -e "\033[32m[OK]\033[0m openpyxl"; (( PASS++ )); } \
|| { echo -e "\033[31m[FAIL]\033[0m openpyxl - pip install openpyxl"; (( FAIL++ )); }
python3 -c "import reportlab" 2>/dev/null \
&& { echo -e "\033[32m[OK]\033[0m reportlab"; (( PASS++ )); } \
|| { echo -e "\033[31m[FAIL]\033[0m reportlab - pip install reportlab"; (( FAIL++ )); }

echo ""
echo "=== Layer 1 – Secrets & AV ==="
check "betterleaks" betterleaks
check "detect-secrets" detect-secrets
check "ClamAV" clamscan
check "YARA" yara

echo ""
echo "=== Layer 2 – OWASP/CWE ==="
check "Semgrep" semgrep

echo ""
echo "=== Layer 3 – SCA/CVE ==="
check "trivy" trivy
check "pip-audit" pip-audit
check "safety" safety
check "npm" npm

echo ""
echo "=== Layer 5 – SAST ==="
check "Bandit" bandit

```

```

check "ShellCheck" shellcheck
check "cppcheck" cppcheck
check "hadolint" hadolint
check "checkov" checkov

echo ""
echo "=== Layer 6 – Licence ==="
check "scancode" scancode

echo ""
echo "=== Summary ==="
echo -e " \033[32mOK: $PASS\033[0m | \033[33mOptional missing: $WARN\033[0m | \033[31mRequired missing: $FAIL\033[0m"
[[ $FAIL -eq 0 ]] \
&& echo -e "\033[32mInstallation COMPLETE – pipeline ready\033[0m" \
|| echo -e "\033[31mInstallation INCOMPLETE – fix FAIL items above\033[0m"

```

```

chmod +x verify_install.sh && ./verify_install.sh

```

10. Air-Gapped Operation

The pipeline supports fully disconnected operation through `--offline`. This section is the reference for what each tool needs in that mode.

10.1 Why `--offline` is required, and not merely advisable

Several scanners reach the network at **scan** time, not just at install time. Without `--offline` on an isolated host those calls are still attempted. They do not crash the run — each is individually guarded — but they block on DNS and TCP timeouts and then return nothing.

The consequence is worse than slowness. Layer 2 (OWASP/CWE) and Layer 3 (CVE) contribute no findings, while the report still presents them as having run. The verdict can be **PASS on a repository that was never meaningfully scanned**.

`--offline` changes that: each tool is pointed at locally staged data and given the flags that suppress update attempts and telemetry, and any layer that genuinely cannot run is recorded as an explicit `OFFLINE`: warning naming what did not execute. A silent gap becomes a stated one.

```

export OFFLINE_CACHE=/opt/ai-transit/offline-cache
./ai_transit.sh --offline /path/to/repo

```

Remote URLs are refused in this mode — copy the repository to the host and pass its path.

10.2 Per-tool reference

Group A — works offline with no preparation. Rules are compiled into the tool; nothing to stage, no flags required.

Tool	Layer	Notes
betterleaks	L1	Detection rules embedded in the binary
detect-secrets	L1	Entropy and regex rules embedded
YARA	L1	Reads your own .yar files from \$WORK_DIR/yara-rules/
grep built-ins	L4	CWE-798/22/918/327/338 patterns, entirely local
Bandit	L5	Local Python AST analysis
ShellCheck	L5	Local
cppcheck	L5	Local
hadolint	L5	Local

Group B — works offline once data is staged. These are the ones that make air-gapped operation worth configuring properly.

Tool	Layer	Must be staged	Flags applied by --offline
Semgrep	L2	One YAML per ruleset in \$SEMGREP_RULES_DIR	--config <file> instead of p/<name>, plus --metrics=off
trivy	L3	Vulnerability DB in \$TRIVY_CACHE_DIR	--skip-db-update --skip-java-db-update --offline-scan --cache-dir
ClamAV	L1	*.cvd / *.cld in \$CLAMAV_DB_DIR	--database=<dir>
checkov	L5	Nothing	--skip-download (stops remote provider-schema fetches)
ScanCode	L6	Nothing for licence/copyright	--vulnerability is dropped ; licence and copyright detection are local and still run

Group C — no offline mode exists. These query a remote advisory service by design and cannot be staged. Under --offline they are not attempted, and each emits an OFFLINE: warning so the gap is on the record.

Tool	Layer	Why	Mitigation
pip-audit	L3	Resolves advisories from PyPI/OSV	Python CVEs covered by the staged trivy DB
safety	L3	Queries the Safety DB service	As above
npm audit	L3	Queries the npm registry endpoint	JS CVEs covered by the staged trivy DB
ScanCode --vulnerability	L6	Queries VulnerableCode	Package CVEs covered by the staged trivy DB

Tool	Layer	Why	Mitigation
GitHub API size check	fetch	Needs api.github.com	Not applicable: --offline accepts local paths only

The single practical consequence: **trivy** is the only dependency-CVE coverage you have offline.** If its database is not staged, Layer 3 contributes nothing.

10.3 Environment variables

Variable	Default	Purpose
OFFLINE	false	Same as --offline
OFFLINE_CACHE	\$WORK_DIR/offline-cache	Root of the staged assets
SEMGREP_RULES_DIR	\$OFFLINE_CACHE/semgrep-rules	Exported ruleset YAML files
TRIVY_CACHE_DIR	\$OFFLINE_CACHE/trivy-db	trivy database cache
CLAMAV_DB_DIR	\$OFFLINE_CACHE/clamav	ClamAV signature files

--offline also exports SEMGREP_SEND_METRICS=off, DO_NOT_TRACK=1, CHECKPOINT_DISABLE=1 and PIP_NO_INDEX=1, so no tool blocks on a telemetry or version-check call.

10.4 Building the cache — on a connected host

```
./prepare_offline_cache.sh /path/to/offline-cache
```

It stages the Semgrep rulesets, the trivy database, the ClamAV signatures and your YARA rules, writes a SHA-256 manifest, and prints a summary of what it could and could not stage.

Expect roughly **300–800 MB**, dominated by the trivy database.

10.5 Transferring and verifying

```
# connected host
tar -czf offline-cache.tar.gz -C /path/to offline-cache
sha256sum offline-cache.tar.gz # note this value

# air-gapped host, after transfer
sha256sum offline-cache.tar.gz # must match
tar -xzf offline-cache.tar.gz -C /opt/ai-transit/
cd /opt/ai-transit/offline-cache && sha256sum --check .cache_manifest.sha256
```

10.6 Running

```
export OFFLINE_CACHE=/opt/ai-transit/offline-cache
./ai_transit.sh --offline /path/to/repo
```

```
# Docker (the wrapper forwards OFFLINE and the cache paths)
OFFLINE=true OFFLINE_CACHE=/opt/ai-transit/offline-cache \
./docker-run.sh --offline /path/to/repo
```

10.7 Confirming a clean offline run

The report should contain **no** OFFLINE: warning about Layer 2 or trivy:

```
grep -o 'OFFLINE:[^]]*' "$WORK_DIR"/reports/report_*.json
```

Warning seen	Meaning	Fix
Layer 2 skipped entirely	No rulesets staged — OWASP/CWE did not run	Re-run <code>prepare_offline_cache.sh</code>
semgrep rulesets not staged (...)	Some rulesets missing — partial coverage	Export the named rulesets
trivy database not staged	No dependency CVE coverage at all	Stage the trivy DB
ClamAV signature database is empty	Malware scan found nothing because it had no signatures	Stage <code>*.cvd</code> files
Python/JavaScript dependency CVE scan unavailable	Expected — see Group C	None; trivy covers this

10.8 Keeping the cache current

Vulnerability data ages. A stale trivy database reports a clean result for CVEs published after it was built, which is the same failure mode as not scanning at all — only harder to notice.

Rebuild and re-transfer the cache on a defined cadence (**weekly** is a reasonable default for the trivy DB and ClamAV signatures; Semgrep rulesets change more slowly). Record the build date alongside the archive so the air-gapped side can tell how old its data is.

11. Sample Scans — Testing the Pipeline

11.1 Quick smoke test (local directory)

Create a small test repository to validate the pipeline end-to-end:

```
#!/usr/bin/env bash
# create_test_repo.sh — creates a clean sample repo that should PASS

mkdir -p /tmp/sample-repo/{src,tests,infra}

# Python source
cat > /tmp/sample-repo/src/app.py << 'EOF'
import os
import sqlite3
```

```
def get_user(db_path: str, username: str) -> list:
    """Retrieve user using parameterised query (safe)."""
    conn = sqlite3.connect(db_path)
    return conn.execute(
        "SELECT * FROM users WHERE name = ?", (username,)
    ).fetchall()

def main():
    db = os.environ.get("DB_PATH", "/tmp/app.db")
    print(get_user(db, "alice"))
EOF

# Shell script
cat > /tmp/sample-repo/src/setup.sh << 'EOF'
#!/usr/bin/env bash
set -euo pipefail
APP_DIR="${APP_DIR:-/opt/app}"
mkdir -p "$APP_DIR"
echo "Setup complete"
EOF

# Requirements file (up-to-date)
cat > /tmp/sample-repo/requirements.txt << 'EOF'
requests==2.32.3
Pillow==10.3.0
EOF

# README
echo "# Sample project" > /tmp/sample-repo/README.md

echo "Test repo created at /tmp/sample-repo"
```

```
bash create_test_repo.sh
./ai_transit.sh /tmp/sample-repo
# Expected: PASS – ZIP produced in ./Good/
```

11.2 FAIL test — embedded secret

```
mkdir -p /tmp/fail-secret

cat > /tmp/fail-secret/config.py << 'EOF'
# Bad practice: hardcoded credential
AWS_ACCESS_KEY_ID = "AKIAIOSFODNN7EXAMPLE"
AWS_SECRET_ACCESS_KEY = "wJalrXUtnFEMI/K7MDENG/bPxRfiCYEXAMPLEKEY"
EOF

./ai_transit.sh /tmp/fail-secret
# Expected: FAIL – betterleaks or detect-secrets detects the AWS keys
# Files moved to /opt/ai-transit/quarantine/
```

```
rm -rf /tmp/fail-secret
```

11.3 FAIL test — SQL injection (OWASP A03)

```
mkdir -p /tmp/fail-sqli

cat > /tmp/fail-sqli/db.py << 'EOF'
import sqlite3

def login(username, password):
    conn = sqlite3.connect("app.db")
    query = "SELECT * FROM users WHERE name='" + username + \
    "'" AND pass='" + password + "'"
    return conn.execute(query).fetchall()
EOF

./ai_transit.sh /tmp/fail-sqli
# Expected: FAIL – Semgrep detects SQL injection (OWASP A03 / CWE-89)

rm -rf /tmp/fail-sqli
```

11.4 FAIL test — vulnerable dependency (CVE)

```
mkdir -p /tmp/fail-cve

cat > /tmp/fail-cve/requirements.txt << 'EOF'
Pillow==9.0.0
requests==2.18.0
EOF

./ai_transit.sh /tmp/fail-cve
# Expected: FAIL – trivy/pip-audit detect CVEs in Pillow 9.0.0

rm -rf /tmp/fail-cve
```

11.5 WARN test — risky licence (GPL)

```
mkdir -p /tmp/warn-licence

cat > /tmp/warn-licence/main.py << 'EOF'
# This file is licensed under the GNU General Public License v3.0
# Copyright (C) 2024 Example Corp.
# You may redistribute and/or modify under the terms of the GPL.

def hello():
    print("hello")
EOF
```

```
./ai_transit.sh /tmp/warn-licence
# Expected: PASS (WARN logged) – ScanCode detects GPL-3.0 licence
# ZIP produced in ./Good/ but WARN logged for legal review

rm -rf /tmp/warn-licence
```

11.6 FAIL test — Dockerfile misconfigurations

```
mkdir -p /tmp/fail-docker

cat > /tmp/fail-docker/Dockerfile << 'EOF'
FROM ubuntu:latest
ENV DB_PASSWORD=mysecretpassword
RUN apt-get update && apt-get install -y curl
ADD https://example.com/app.tar.gz /app/
EOF

./ai_transit.sh /tmp/fail-docker
# Expected: FAIL – hadolint detects :latest tag, ENV secret, ADD from URL

rm -rf /tmp/fail-docker
```

11.7 Scan a real public GitHub repository

```
# Scan a small, well-known public repo (adjust URL as needed)
./ai_transit.sh https://github.com/psf/requests main
# Expected: PASS or WARN – clean repo, dependencies may have minor CVEs
```

11.8 Run the self-check to verify the pipeline itself

```
# PDF report (default)
python3 selfcheck.py --bundle-dir . --output selfcheck_report.pdf

# JSON report (machine-readable, CI-friendly)
python3 selfcheck.py --bundle-dir . --format json --output selfcheck_report

# Both formats at once
python3 selfcheck.py --bundle-dir . --format both --output selfcheck_report

# Run only specific checks (faster)
python3 selfcheck.py --bundle-dir . --only 11.1,11.4,11.6

# Expected output:
# §11.1 Meta-scan → PASS or WARN
# §11.2 Binary checks → SKIP (unless --checksums provided)
# §11.3 GPG/cosign → PASS or WARN
# §11.4 Python CVE → PASS
# §11.5 Host OS CVE → PASS or WARN
```

```
# §11.6 Bundle integrity → PASS
# §11.7 AIDE → SKIP (unless AIDE installed)
```

11.9 CI mode — quiet verdict with severity filter

```
# Only block on CRITICAL CVEs/findings; warnings and lower are ignored
./ai_transit.sh --quiet --min-severity critical https://github.com/org/repo
echo "Exit code: $?" # 0 = PASS, 1 = FAIL
```

11.10 Diff mode — scan only changed files (PR workflow)

```
# Scan only files changed since the merge base commit
SINCE=$(git merge-base HEAD origin/main)
./ai_transit.sh --since "$SINCE" /path/to/local/repo
```

11.11 Private repository scan

```
# Generate a fine-grained PAT with "Contents: read" on github.com
export GITHUB_TOKEN="ghp_your_token_here"
./ai_transit.sh https://github.com/myorg/private-repo main
```

11.12 Report-only mode (audit without blocking)

```
# Scan and generate reports but always exit 0 – useful for first-pass auditing
./ai_transit.sh --report-only https://github.com/org/repo
# Reports are written to $WORK_DIR/reports/ regardless of findings
```

11.13 Air-gapped scan

```
# On a connected host: build the cache and transfer it
./prepare_offline_cache.sh /tmp/offline-cache
tar -czf offline-cache.tar.gz -C /tmp offline-cache

# On the air-gapped host
tar -xzf offline-cache.tar.gz -C /opt/ai-transit/
cd /opt/ai-transit/offline-cache && sha256sum --check .cache_manifest.sha256

export OFFLINE_CACHE=/opt/ai-transit/offline-cache
./ai_transit.sh --offline /path/to/repo

# Confirm no layer was silently skipped
grep -o 'OFFLINE:[^]]*' "$WORK_DIR/reports/report_*.json
# Expected: only the Python/JavaScript dependency notices (see 10.2 group C)
```

12. Self-Scan: Verifying the Installation is Safe

Before deploying in a production environment, run the full self-check:

```
python3 selfcheck.py --bundle-dir . --output selfcheck_report.pdf
```

This executes all §11 checks (meta-scan, binary checksums, GPG/cosign, Python CVE scan, host OS CVE, bundle integrity, AIDE) and produces a colour-coded PDF report. See `selfcheck.py --help` for all options.

12.1 Output formats and selective checks

```
python3 selfcheck.py --format json --output selfcheck_report # machine-readable
python3 selfcheck.py --format both --output selfcheck_report # PDF + JSON
python3 selfcheck.py --only 11.1,11.4,11.6 # subset, faster
```

The JSON report carries a top-level verdict field (PASS / WARN / FAIL), which is what a CI job should key on.

12.2 The bundle manifest — generate it at install time

Check §11.6 (bundle file integrity) compares every bundle file against `.bundle_manifest.sha256`. That manifest is **deliberately not tracked in version control**: if it were, it would report "File tampering detected" after every ordinary edit, and a check that cries wolf is a check people learn to ignore. Git already guarantees the integrity of the repository; the manifest's job is to detect tampering with an **installed** bundle.

Generate it once, immediately after installing and verifying the bundle:

```
python3 selfcheck.py --write-manifest
```

Regenerate it after any *intentional* change to the bundle. From then on, any §11.6 failure means a file changed without your knowledge.

```
# Verify (should PASS on an untouched installation)
python3 selfcheck.py --only 11.6
```

The manifest stores **relative** paths, so it stays valid if the bundle is moved or installed to a different prefix. Verification must therefore be run from the bundle directory (`selfcheck.py` does this automatically).

13. Security Hardening Recommendations

Dedicated service account

```
sudo useradd -r -m -d /opt/ai-transit -s /bin/bash aitransit
sudo chown -R aitransit:aitransit /opt/ai-transit
```

```
sudo -u aitransit ./ai_transit.sh https://github.com/org/repo
```

Network isolation (allow only github.com outbound)

```
sudo ufw default deny outgoing
sudo ufw allow out 443 comment "HTTPS for GitHub"
sudo ufw allow out 53 comment "DNS"
sudo ufw enable
```

Quarantine filesystem (noexec)

```
# /etc/fstab
tmpfs /opt/ai-transit/quarantine tmpfs rw,noexec,nosuid,nodev,size=2G 0 0
```

Weekly tool updates (cron)

```
# /etc/cron.weekly/ai-transit-update
#!/bin/bash
freshclam
trivy image --download-db-only
source /opt/ai-transit/venv/bin/activate
pip install --upgrade betterleaks detect-secrets semgrep bandit \
pip-audit safety checkov scancode-toolkit
```

Log rotation

```
# /etc/logrotate.d/ai-transit
/opt/ai-transit/logs/*.log {
daily
rotate 30
compress
missingok
notifempty
}
```

14. Troubleshooting

Symptom	Likely cause	Fix
betterleaks: command not found	Binary not in PATH	<code>sudo cp ~/go/bin/betterleaks /usr/local/bin/</code>
freshclam: connect refused	clamav-freshclam not running	<code>sudo systemctl start clamav-freshclam</code>
semgrep: network error	No internet — first run	Pre-download rules (see §10)
trivy: DB not found	First run without internet	<code>trivy image --download-db-only</code>

Symptom	Likely cause	Fix
scancode: takes too long	Large repo	Use <code>--timeout 30</code> to limit per-file time
pip-audit: no vulnerabilities on old packages	DB unreachable	Check internet; use trivy offline instead
npm audit: ENOLOCK	No package-lock.json	Run <code>npm install --package-lock-only</code> first
openpyxl manquant	Not installed in venv	<code>pip install openpyxl</code>
declare -A: invalid option	Bash < 4.0	Install bash 5: <code>brew install bash</code> / use WSL2
zip: command not found	zip not installed	<code>sudo apt-get install zip</code>
WARN on all files	Missing optional tools	Install missing tools; WARNs do not block
Git clone fails on a private repo	No credentials	Set <code>GITHUB_TOKEN</code> (see §8.2); the token is passed via <code>GIT_ASKPASS</code> , never in the clone URL
Repository not found or private (HTTP 404)	Private repo without a token, or a typo	Set <code>GITHUB_TOKEN</code> , or check the URL
§11.6 reports "File tampering detected" after your own edits	Manifest is stale	Regenerate it: <code>python3 selfcheck.py --write-manifest</code> (see §12.2)
--only reports 0 checks and exits 2	Check IDs mistyped (1.1 instead of 11.1)	Use the full IDs: 11.1 ... 11.7
Findings appear against the wrong file	Report written by a pre-P8 version	Upgrade; the JSON writer dropped empty records and shifted rows
Allowlist entries have no effect	.transit-allow.json not at the repository root, or rule/path do not match	path is relative to the repo root; rule matches the finding's leading token, e.g. CWE-89
Diff mode scans everything	--since commit unreachable in a shallow clone	The pipeline warns and falls back to a full scan; fetch more history or use a local path
Docker build fails downloading trivy	Pinned version no longer published	Update <code>ARG TRIVY_VERSION / TRIVY_SHA256</code> (see §17.1); the pins CI job prints the correct digest
ZIP contains tmp/.../fetch/repo_... paths	Archive built by a pre-P8 version	Upgrade; archives are now rooted at the repository
Scan hangs for minutes on an isolated host	Running without --offline; tools are blocking on network timeouts	Use --offline (see §10)
Offline run passes suspiciously fast with few findings	Layers 2 and 3 had no staged data	Check for OFFLINE: warnings in the report; stage the cache (§10.4)
OFFLINE:Layer 2 skipped entirely	Semgrep rulesets not staged	Run <code>prepare_offline_cache.sh</code> on a connected host
OFFLINE:trivy database not staged	No dependency CVE coverage offline	Stage the trivy DB; it is the only offline CVE source

15.2 Fixtures

tests/fixtures/ holds small repositories, each with one job:

Fixture	Expected
clean/	PASS
vulnerable/	FAIL
allowlisted/	PASS – the finding is downgraded by .transit-allow.json
ignored/	PASS – the offending file is excluded by .transitignore
rules/	A corpus where each file triggers, or deliberately does not trigger, one rule

rules/safe_sql.py is a regression guard: it contains correct parameterised queries that a previous version of the SQL rule wrongly flagged as injection.

15.3 Adding a rule

Add **both** a file that must trigger the rule and a similar-but-safe file that must not, then confirm the new assertion **fails before the rule exists**. A test that has never been observed failing proves nothing — during development of this suite, two tests passed against known-broken code because the tests themselves were wrong.

16. Continuous Integration

.github/workflows/ci.yml runs on every push and pull request.

Job	Purpose	Blocking
lint	shellcheck (errors fatal, warnings advisory) + Python syntax	Yes
test	Test suite with no scanning tools — the fast signal	Yes
test-with-tools	Suite again with semgrep/bandit/detect-secrets installed; exercises tool-dependent paths the degraded run cannot reach	No
pins	Downloads each pinned tool artifact, fails if the version does not exist, and compares against the pinned SHA-256	Yes
docker	Builds the image, asserts it runs as non-root, smoke-tests both fixtures	Yes

The docker job is the only place the multi-stage build, the pinned tool versions and the non-root user are actually exercised — none of that can be verified by the test suite alone.

If `pins` fails, the pinned version in the `Dockerfile` no longer resolves to a downloadable artifact. The job log prints the correct SHA-256 for the current pin; update `ARG TRIVY_VERSION` / `ARG TRIVY_SHA256` accordingly.

17. Docker Image — Build Arguments & Integrity

The image is a two-stage build: a builder stage compiles `betterleaks` with Go, and the runtime stage copies only the resulting binary, so the Go toolchain never reaches the final image. It runs as the non-root user `transit`.

17.1 Build arguments

Argument	Default	Purpose
<code>TRIVY_VERSION</code>	<code>0.58.2</code>	trivy release to install
<code>TRIVY_SHA256</code>	<i>(empty)</i>	SHA-256 of the trivy tarball; empty disables verification
<code>HADOLINT_VERSION</code>	<code>2.12.0</code>	hadolint release to install
<code>HADOLINT_SHA256</code>	<code>56de6d5e...</code>	SHA-256 of the hadolint binary (verified)
<code>BETTERLEAKS_VERSION</code>	<code>0.1.0</code>	betterleaks module version

```
docker build -t ai-transit:latest \  
--build-arg TRIVY_VERSION=0.58.2 \  
--build-arg TRIVY_SHA256=<digest> .
```

17.2 Why the digests matter

Pinning a version defends against getting a *different release*. It does not defend against getting a *different binary* for that release — a compromised or replaced artifact keeps the same version string. The digest closes that gap.

When a digest argument is empty the build still succeeds, but prints a warning **and the artifact's actual digest**, so the correct value can be pasted in without a separate download. The `pins` CI job prints the same value.

TRIVY_SHA256 ships empty because the pinned trivy version could not be resolved from the development environment. Fill it in from the first `pins` CI run before treating the image as production-ready.

17.3 Running through the wrapper

`docker-run.sh` forwards pipeline flags into the container, so the Docker and native interfaces behave identically:

```
./docker-run.sh --build # build once  
./docker-run.sh --quiet --min-severity critical https://github.com/org/repo
```

```
./docker-run.sh --report-only /path/to/local/repo
```

Environment variables (WORK_DIR, MAX_SIZE_MB, MIN_SEVERITY, VERBOSITY, GITHUB_TOKEN, SINCE_COMMIT) are forwarded automatically when set.

A GITHUB_TOKEN passed to a container is visible via `docker inspect` to anyone who can reach the Docker daemon. On a shared host, prefer running the pipeline natively for private repositories.

Quick Reference — One-Line Install (Ubuntu 22.04 / 24.04)

```
# Step 1: system packages
sudo apt-get update && sudo apt-get install -y \
bash git curl jq zip unzip file coreutils wget gpg \
python3 python3-pip python3-venv build-essential golang-go \
nodejs npm shellcheck cppcheck clamav clamav-daemon yara

# Step 2: Python venv + packages
python3 -m venv /opt/ai-transit/venv
source /opt/ai-transit/venv/bin/activate
pip install --upgrade pip
pip install openpyxl reportlab detect-secrets bandit pip-audit safety \
semgrep checkov scancode-toolkit

# Step 3: betterleaks
go install github.com/betterleaks/betterleaks@latest
sudo cp ~/go/bin/betterleaks /usr/local/bin/betterleaks

# Step 4: trivy
curl -sL https://raw.githubusercontent.com/aquasecurity/trivy/main/contrib/install.sh \
| sudo sh -s -- -b /usr/local/bin

# Step 5: hadolint
VERSION=$(curl -s https://api.github.com/repos/hadolint/hadolint/releases/latest \
| jq -r '.tag_name')
sudo curl -sSL \
"https://github.com/hadolint/hadolint/releases/download/${VERSION}/hadolint-Linux-x86_64" \
-o /usr/local/bin/hadolint && sudo chmod +x /usr/local/bin/hadolint

# Step 6: ClamAV DB update
sudo freshclam

# Step 7: trivy DB
trivy image --download-db-only

echo "Installation complete – run ./verify_install.sh to confirm"
```